Fluid Mechanics ME 21201 Quiz 2

September 11, 2026

1. (2 pts) Consider a steady, two-dimensional inviscid flow in the streamline coordinate system $(\boldsymbol{s}, \boldsymbol{n})$, where $\boldsymbol{s}$ is along the streamline and $\boldsymbol{n}$ points toward the center of curvature with local radius R. The normal component of the Euler equation in the absence of body forces gives the transverse pressure gradient as $\frac{\partial p}{\partial \boldsymbol{n}} = \frac{\rho \boldsymbol{v}^2}{R}$. If the velocity field is also irrotational, how must the spatial derivative of flow speed along the normal direction, $\frac{\partial v}{\partial \boldsymbol{n}}$, relate to $\boldsymbol{v}$ and R?

   A. $\frac{\partial \boldsymbol{v}}{\partial \boldsymbol{n}} = \frac{\boldsymbol{v}}{R}$
   B. $\frac{\partial \boldsymbol{v}}{\partial \boldsymbol{n}} = -\frac{\boldsymbol{v}}{R}$
   C. $\frac{\partial \boldsymbol{v}}{\partial \boldsymbol{n}} = \frac{2\boldsymbol{v}}{R}$
   D. $\frac{\partial \boldsymbol{v}}{\partial \boldsymbol{n}} = 0$

   $\omega_z = -\frac{\partial \boldsymbol{v}}{\partial \boldsymbol{n}} + \frac{\boldsymbol{v}}{R}$

2. (2 pts) Kelvin's circulation theorem states that circulation Γ is conserved along a closed material loop in an inviscid fluid under conservative body forces. When the fluid is baroclinic ($\nabla p \times \nabla \rho \neq 0$), what is the expression for the material derivative of circulation $\frac{D\Gamma}{Dt}$?

   A. $\oint_C (\rho \nabla \rho \times \nabla p) \cdot \mathrm{d}\boldsymbol{l}$
   B. $\iint_S (\rho \nabla \rho \times \nabla p) \cdot \boldsymbol{n} \mathrm{d}A$
   C. $\iint_S \frac{1}{\rho^2} (\nabla \rho \times \nabla p) \cdot \boldsymbol{n} \mathrm{d}A$
   D. -$\iint_S \frac{1}{\rho^2} (\nabla \rho \times \nabla p) \cdot \boldsymbol{n} \mathrm{d}A$

   $\iint_S \frac{1}{\rho^2} (\nabla \rho \times \nabla p) \cdot \boldsymbol{n} \mathrm{d}A.$

3. (1 pt) For a two-dimensional compressible planar flow of a fluid with a spatially and temporally varying density $\rho(x, y, t)$, a stream function $\psi(x, y, t)$ can be formulated to identically satisfy the 2D continuity equation. How are the velocity components u and v related to ψ?.

   A. $u = \frac{\partial \psi}{\partial y}$, $v = -\frac{\partial \psi}{\partial x}$
   B. $u = \frac{\partial \psi}{\partial x}$, $v = -\frac{\partial \psi}{\partial y}$
   C. $\rho u = \frac{\partial \psi}{\partial x}$, $\rho v = -\frac{\partial \psi}{\partial y}$
   D. $\rho u = \frac{\partial \psi}{\partial y}$, $\rho v = -\frac{\partial \psi}{\partial x}$

   $\rho u = \frac{\partial \psi}{\partial y}$, $\rho v = -\frac{\partial \psi}{\partial x}$

4. (2 pts) While a single stream function ψ works perfectly for 2D planar flows, a 3D incompressible flow requires a vector potential or dual stream functions. If a flow field is described by two independent scalar functions $f(x, y, z)$ and $g(x, y, z)$, such that the velocity vector is represented as $\boldsymbol{u} = \nabla f \times \nabla g$, which of the following statements is true regarding this formulation?

A. The flow automatically satisfies the irrotational condition.

B. The flow automatically satisfies the incompressibility condition.

C. The lines of intersection of the surfaces $f =$ const and $g =$ const are perpendicular.

D. The functions f and g must satisfy $\nabla^2 f = \nabla^2 g = 0$.

Using the vector identity for divergence of a cross product: $\nabla \cdot (\nabla f \times \nabla g) = \nabla g \cdot (\nabla \times \nabla f) - \nabla f \cdot (\nabla \times \nabla g)$. Since the curl of any gradient is identically zero, the divergence is automatically zero, ensuring mass conservation for an incompressible fluid.

5. (1 pt) The velocity gradient tensor $\nabla \boldsymbol{u}$ can be decomposed into a symmetric strain-rate tensor $\boldsymbol{S}$ and an anti-symmetric rotation tensor $\boldsymbol{\Omega}$. Consider a steady, 2D velocity field where the velocity components are given by $u = Ax + By$ and $v = Cx + Dy$. If the flow is both incompressible and irrotational, what constraint must be placed on the constants A, B, C, and D?

A. $A = D$, $B = C$

B. $A = D$, $B = -C$

C. $A = -D$, $B = C$

D. $A = -D$, $B = -C$

$A = -D$, $B = C$

6. (2 pts) Consider a two-dimensional, unsteady velocity field given by $u = U_0$, $v = kt$, where U_0 and k are positive constants, and t is time. At any given instant $t = t_0$, what is the relationship between the slope of the streamline passing through the origin and the slope of the pathline of a particle that passed through the origin at time $t = 0$?

A. They are identical.

B. The streamline slope is exactly twice the pathline slope.

C. The streamline slope is exactly half the pathline slope.

D. None of these.

They are identical.